

Service - CSV Combination API Processor

This folder contains the CSV Combination API Processor service that generates all possible combinations of industry and location pairs from a CSV file, makes API calls, and saves the results.

Files Overview

Core Scripts

- **combination_api_processor.py** - Main processor script with resource control
- **run_combination_processor.py** - Python wrapper script for easy execution
- **test_setup.py** - Setup verification test script

Configuration

- **config_combination_processor.json** - Configuration settings
- **requirements_combination_processor.txt** - Python dependencies

Execution Scripts

- **run_combination_processor.sh** - Linux/Mac shell script
- **run_combination_processor.bat** - Windows batch script

Documentation & Examples

- **README_combination_processor.md** - Detailed documentation
- **sample_input.csv** - Example input file format

Quick Start

Linux/Mac

```
# Make script executable (if needed)
chmod +x run_combination_processor.sh

# Run the processor
./run_combination_processor.sh

# Or check setup only
./run_combination_processor.sh --check

# Or install dependencies only
./run_combination_processor.sh --install

# Or run setup test
./run_combination_processor.sh --test
```

Windows

```
# Double-click or run from command prompt
run_combination_processor.bat
```

Python Direct

```
# Install dependencies
pip install -r requirements_combination_processor.txt

# Run processor
python run_combination_processor.py
```

Configuration

Edit `config_combination_processor.json` to customize:

- API endpoint URL
- Resource limits (memory, CPU)
- Processing parameters (workers, rate limiting)
- CSV column names

Input Format

Your CSV should have these columns:

```
industry,city,state,country
software development,Dallas,TX,USA
healthcare,Los Angeles,CA,USA
```

Output Files

The service generates:

- `results_output.csv` - Main results
- `results_output_errors.csv` - Failed API calls
- `combination_processor.log` - Processing logs
- `results_output_batch_N.csv` - Intermediate results

Resource Control

The service automatically:

- Monitors memory and CPU usage
- Pauses when resources are high
- Limits API call rate
- Processes in manageable batches

Detailed Documentation

See [README_combination_processor.md](#) for complete documentation including:

- Advanced usage examples
- Troubleshooting guide
- Performance optimization tips
- API requirements

Requirements

- Python 3.7+
- pandas, requests, psutil
- Running API server (default: localhost:5000)

Service Integration

This service is designed to run alongside your main application with minimal resource impact. It can be:

- Run as a background service
- Scheduled for batch processing
- Integrated into larger workflows
- Monitored via log files